

**Cross-Subject Generalization in EEG-Based Covert Speech Decoding: A Systematic
Review and Governance Implications**

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Abstract

Cross-subject generalization, the ability of a decoding model to classify EEG signals from individuals absent from training, is the central unsolved challenge in EEG-based imagined speech brain-computer interfaces and the primary barrier to deployable assistive communication technology for individuals with severe motor impairments. This systematic review examined the current state of cross-subject generalisation in EEG-based covert speech decoding and the governance implications that follow from identified limitations. Following PRISMA 2020 guidelines, a search across four databases identified nine studies meeting a strict operational definition of cross-subject generalisation requiring complete subject-level exclusion from all stages of model training and evaluation. Verified zero-shot cross-subject performance clustered between 29% and 32% above a 25% chance baseline on the most common benchmark dataset, regardless of generalisation strategy. Higher figures in the literature were attributable to binary tasks, calibration dependency, or evaluation designs not reflecting deployment conditions. Approaches achieving higher accuracy consistently required more target subject data, while zero-calibration approaches remained at or marginally above chance. A recurring methodological concern was that several studies described cross-subject evaluations without clearly implementing strict subject-level holdout procedures. Clinically meaningful zero-shot generalisation has not yet been demonstrated in this field. The inverse relationship between performance and calibration burden poses a structural accessibility problem for the populations this technology is designed to serve, and governance frameworks addressing evaluation standards, reporting requirements, and equitable access are needed before clinical translation proceeds.

To the researchers, clinicians, and patients who together make this field worth pursuing.

And to the possibility that one day, thought alone will be enough.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

ALS – Amyotrophic Lateral Sclerosis

APSIPA ASC – Asia Pacific Signal and Information Processing Association Annual Summit and Conference

BCI – Brain-Computer Interface

CAIP – Computer Analysis of Images and Patterns

CCV – Channel Cross-Covariance

CNN – Convolutional Neural Network

CSP – Common Spatial Pattern DeepConvNet – Deep Convolutional Network

DenseNet – Densely Connected Convolutional Network

DT – Decision Tree

DWT – Discrete Wavelet Transform

ECoG – Electrocorticography

EEG – Electroencephalography

EEGNet – Electroencephalography Network

ELM – Extreme Learning Machine

FDA – Food and Drug Administration

FFT – Fast Fourier Transform

FIR – Finite Impulse Response

fMRI – Functional Magnetic Resonance Imaging

fNIRS – Functional Near-Infrared Spectroscopy

GAN – Generative Adversarial Network

GDPR – General Data Protection Regulation

GRU – Gated Recurrent Unit

HiFiGAN – High Fidelity Generative Adversarial Network

HuBERT – Hidden Unit Bidirectional Encoder Representations from Transformers

ICA – Independent Component Analysis

ICASSP – International Conference on Acoustics, Speech and Signal Processing

IoT – Internet of Things

LDA – Linear Discriminant Analysis

LOSO – Leave-One-Subject-Out

LSTM – Long Short-Term Memory

MAML – Model-Agnostic Meta-Learning

MEG – Magnetoencephalography

PCA – Principal Component Analysis

PRISMA – Preferred Reporting Items for Systematic Reviews and Meta-Analyses

ResNet – Residual Network

RF – Random Forest

SNR – Signal-to-Noise Ratio

SVM – Support Vector Machine

TCN – Temporal Convolutional Network

t-SNE – t-Distributed Stochastic Neighbor Embedding

VAE – Variational Autoencoder

XGBoost – Extreme Gradient Boosting

Cross-Subject Generalization in EEG-Based Covert Speech Decoding: A Systematic Review and Governance Implications

"Speech is the mirror of the soul; as a man speaks, so he is." - Publilius Syrus, Moral Sayings, 1st century BC.

For individuals living with ALS, locked-in syndrome, or severe motor impairment, that mirror breaks while the neural substrates of language remain intact. The capacity to generate speech survives; the motor pathway to externalize it does not. Imagined speech, the internal rehearsal of words and sentences without any audible or visible output, offers a way to bridge that gap. The neural signatures it produces are, in principle, decodable, offering the possibility of a communication channel that requires nothing from the user except thought itself.

Brain-computer interfaces, and in particular EEG-based systems, have emerged as the practical vehicle for pursuing this possibility, combining the non-invasive accessibility that real-world deployment demands with a growing body of machine learning approaches capable of extracting meaningful signals from the complex, noisy landscape of human neural activity. Yet as these systems grow in sophistication, a question that is both technically and ethically urgent has come into sharper focus: how well do decoding models generalise across individuals, and what does the answer to that question mean for the populations this technology is meant to serve? Cross-subject generalisation, thus, sits at the intersection of the field's technical ambitions with its clinical promises, and how the literature has addressed it, with what methods, under what assumptions, and with what silences, is the central concern of this review.

The Neural Basis of Imagined Speech

An understanding of the neural substrates underlying imagined speech is a prerequisite for understanding what EEG-based decoding systems are, in practice, attempting to capture.

Imagined speech, along with its near-synonyms inner speech, silent speech, and covert speech, which are used interchangeably across the literature (Zivin, 1979), aims to synthesize audible speech from neural signals without actual vocalization or the use of articulators, providing a more direct bridge to communication for individuals who are unable to vocalize due to physical limitations. A broader constellation of terms including subvocal speech, auditory imagery, internal dialogue, and self-verbalization also appears across disciplines (Morin & Michaud, 2007), though with less terminological consistency and often with subtly different theoretical connotations. Beyond neuroscience, imagined speech decoding holds promise with broader engineering domains, from information security and authentication (Wang et al., 2021) to IoT-based device control (Lee et al., 2022), signalling its emergence as a technology with relevance well outside the clinical context.

The cortical regions most consistently implicated in imagined speech span all four major lobes, each contributing distinct functions to the language production chain. The frontal lobe, and specifically Broca's area in the left inferior frontal gyrus, is responsible for the planning and coordination of verbal expression (Hickok, 2009; Kimberg & Farah, 1993). The temporal lobe houses Wernicke's area in the posterior superior region of the left hemisphere, which processes incoming linguistic content and converts it into semantically meaningful representations (Binder, 2015; Rauschecker & Scott, 2009). These two regions are structurally connected by the arcuate fasciculus, a white matter tract that serves as the primary conduit for information flow between semantic comprehension and speech output stages (Bernal & Ardila, 2009). The parietal lobe contributes to spatial processing and action control (Culham & Kanwisher, 2001), while the occipital lobe, though primarily visual, plays a supporting role in the broader perceptual network engaged during language tasks (Goodale & Milner, 1992).

The relationship between imagined and overt speech is best understood through the dual-stream model of speech processing, which organises language function into two parallel pathways (Hickok & Poeppel, 2007). The dorsal stream handles sensorimotor integration and phonological encoding via the arcuate fasciculus; the ventral stream supports semantic comprehension through temporal-frontal connections. Imagined speech predominantly engages the dorsal stream but diverges from overt speech in ways consequential for EEG-based decoding. Where overt speech is dominated by broadband high-frequency activity localised to sensorimotor and superior temporal regions, reflecting motor execution and auditory feedback, imagined speech relies more heavily on low-frequency oscillations and cross-frequency coupling, with attenuated high-frequency contributions owing to the absence of articulatory movement and auditory self-monitoring (Gross et al., 2013; Hovsepyan et al., 2020). Spatially, imagined speech engages bilateral temporal regions with reduced superior temporal gyrus activity, in contrast to the strong left-lateralised sensorimotor dominance of overt speech, reflecting a shift from motor-driven dynamics toward more distributed, abstract linguistic representation (Gajardo-Vidal et al., 2021; Wilson et al., 2020).

Neural activity associated with speech has been investigated across a range of neuroimaging modalities, including electroencephalography (EEG), functional near-infrared spectroscopy (fNIRS), functional magnetic resonance imaging (fMRI), magnetoencephalography (MEG), and electrocorticography (ECoG), each offering a distinct trade-off between spatial and temporal resolution. However, EEG stands out as the predominant choice owing to its inherent non-invasiveness, portability, and high temporal resolution.

EEG studies have sought to localise imagined speech activations using the internationally standardised 10-10 electrode placement system, which maps recordings onto frontal, temporal,

parietal, central, and occipital regions. Frontal and temporal electrodes, overlying Broca's and Wernicke's areas, consistently carry the most discriminative signal (Bisla & Anand, 2023), with a left-hemisphere dominance reflecting the well-established lateralisation of language function (Biswas & Sinha, 2018). Features such as phase (Bisla & Anand, 2023; Panachakel et al., 2021a; Panachakel et al., 2021b), phase locking value (Lee et al., 2021), and spatio-temporal patterns (Fitriah et al., 2023; Lee et al., 2021; Preedapirat & Wongsawat, 2022) have been used to characterise these activations, with phase locking value showing particular sensitivity in both areas during imagined speech tasks.

Concluded from the above, imagined speech can be precisely seen as a distributed, low-frequency dominated, and left-lateralised process that shares broad architecture with overt speech but lacks its strong motor and auditory anchors, producing a neural signature that is simultaneously subtle, highly variable across individuals, and the precise target that EEG-based decoding systems must learn to read.

Data Acquisition, Preprocessing, and Decoding

Translating the neural signatures of imagined speech into a functional decoding pipeline demands careful methodological attention at every stage, from signal acquisition to final classification. Each step introduces design decisions that fundamentally shape what a decoder can and cannot learn, and a precise understanding of these decisions is essential groundwork for evaluating the cross-subject generalisation problem that this review takes as its central concern.

The experimental setup for EEG-based imagined speech recording involves several interdependent choices. Electrode placement follows internationally standardised systems, most commonly the 10-10 or 10-20 configurations, which ensure consistency across subjects and recording sites (Sharbrough et al., 1991). High-density caps with 64 or 128 channels are typical

in research settings, though studies have shown that carefully selected subsets of 16 or 32 electrodes, concentrated in language-relevant regions, can preserve much of the decoding performance at reduced complexity (Li et al., 2021). The content of the speech imagery task also varies considerably across studies, with the majority targeting phonemes, syllables, or individual words rather than full sentences, reflecting both the difficulty of the decoding task and the structure of available public datasets (K.-W. Lee et al., 2022; J.-W. Lee et al., 2023; Sharon et al., 2020). Stimulus modality ranges from auditory prompts (Sakai et al., 2021) to text and image-based cues (Akshi & Rao, 2021; LaRocco et al., 2023; Sree & Kavitha, 2017), and the timing and randomisation of trials is an important design consideration, as consecutive trials for the same target can introduce temporal artifacts that artificially inflate performance (Porbadnigk et al., 2009).

A small number of publicly available datasets have come to anchor much of the field's empirical work. These datasets vary substantially in the number of subjects, sampling rates, channel counts, and the nature of the speech stimuli recorded, ranging from basic vowel phonemes to directional command words (Jin et al., 2025). Most contain fewer than 20 subjects, and the largest publicly available dataset comprises 27 participants, a scale that places meaningful constraints on what cross-subject analyses can reliably demonstrate. The concentration of the literature around a small number of these datasets, particularly those with directional word stimuli, is a structural feature of the field with direct implications for how generalisable reported findings actually are.

Raw EEG signals require substantial preprocessing before they are suitable for decoding. The standard pipeline begins with resampling to harmonise signals across recording setups (Luck, 2014), followed by bandpass filtering, typically using Butterworth (Mahapatra & Bhuyan,

2023) or Chebyshev (Rekrut et al., 2021) filters across a range of approximately 0.5 to 100 Hz, to retain the neural frequency bands relevant to cognition while suppressing noise. Notch filters at 50 or 60 Hz address power line interference (Dekker et al., 2023; Nguyen et al., 2018).

Artifact removal is a central concern given the susceptibility of EEG to contamination from eye movements, muscle activity, and cardiac signals. Independent Component Analysis is the most widely used approach, decomposing the signal into independent sources from which artifactual components can be identified and removed (Cooney et al., 2022; Mahapatra & Bhuyan, 2022a; Mahapatra & Bhuyan, 2022b). Re-referencing, whether to a common average or mastoid electrodes, further reduces noise shared across channels (Hernández-Del-Toro et al., 2021; Kamble et al., 2023; Moctezuma et al., 2019). In the spatial dimension, channel selection methods ranging from neurophysiologically motivated region-based approaches (Agarwal et al., 2020; Idrees & Farooq, 2016; Wu & Chen, 2020) to algorithm-driven feature selection (Datta et al., 2021; Juyal et al., 2023) are used to focus the signal on the electrode sites most informative for the decoding task (Jin et al., 2025). In the temporal dimension, time-window segmentation identifies the periods of neural response most likely to carry discriminative information, with evidence suggesting that the window between approximately 175 and 375 milliseconds post-stimulus carries the most consistent signal (García-Salinas et al., 2023; Varshney & Khan, 2021).

Feature extraction reduces the high-dimensional preprocessed signal into representations that decoding models can learn from. The most commonly used features fall into time-domain statistical measures, frequency-domain representations including Discrete Wavelet Transform and Fast Fourier Transform, spatial features such as Common Spatial Pattern, and nonlinear measures including entropy and fractal dimensions. Each family of features involves tradeoffs:

time-domain statistics are computationally efficient but noise-sensitive, spatial features like CSP exploit inter-channel relationships effectively but may not transfer well across subjects, and nonlinear features capture complex signal dynamics at the cost of interpretability (Jin et al., 2025). Beyond these conventional approaches, more recent work has introduced tailored feature extraction methods including Riemannian geometric representations of covariance matrices (Appriou et al., 2018; Chu et al., 2020; Georgiadis et al., 2022; Li et al., 2020; Pérez et al., 2024), model-oriented feature assembly that reshapes EEG signals into inputs compatible with pre-trained deep learning architectures (Hemakom et al., 2016; Sarmiento et al., 2021), and graph signal processing approaches that account for the spatial topology of electrode networks (García-Salinas et al., 2019).

Decoding approaches have shifted from classical machine learning classifiers applied to hand-engineered features toward end-to-end deep learning architectures, including CNNs, LSTMs, and transformers, that learn spatial and temporal representations directly from the signal, yielding improved within-subject performance. This shift has, however, could even have compounded the cross-subject generalisation problem, as deep models trained on one individual's data tend to encode subject-specific patterns that resist transfer to new individuals, and it is precisely at this juncture that the central hope concern of this review begins.

The decoding model landscape in imagined speech research was divided by Jin et al, (2025) broadly into shallow machine learning methods and deep learning approaches, with the balance between them reflecting the persistent challenge of limited data as much as any principled methodological preference. Classical machine learning approaches to imagined speech decoding have been dominated by SVM, LDA, RF, and DT classifiers, with comparative studies consistently finding SVM and RF among the strongest performers across classification tasks

(Morooka et al., 2018; Hossain et al., 2023). Beyond direct application, several studies refined these methods: notable improvements include Bayesian-optimised SVM variants (Nguyen et al., 2018), RF combined with fuzzy inference for speed-accuracy balance (Torres-García et al., 2016), graph learning leveraging electrode spatial distribution (Einizade et al., 2022), and ensemble ELM with random rotation pruning for classifier diversity (Pawar & Dhage, 2023). Across this body of work, ensemble and kernel-based methods consistently outperformed standard linear classifiers on multiclass imagined speech tasks.

Deep learning methods have progressively taken a larger share of the literature, and CNN has become the dominant deep learning architecture for imagined speech decoding, with convolutional layers enabling extraction of local spatial and frequency patterns from EEG signals. Architecturally, studies have explored multi-dimensional CNN integrating frequency, space, and time (Ko et al., 2021); hybrid spatio-temporal dilated networks (Li et al., 2021); multi-receptive field CNN for high-dimensional feature capture (Park & Lee, 2023); and multi-channel CNN respecting regional spatial-spectral-temporal structure (Neubig & Sellami, 2019). Since EEG signals carry strong temporal dependencies, CNN has frequently been combined with LSTM (Saha & Fels, 2019; Kumar & Scheme, 2021), Bidirectional GRU (Hernandez-Galvan et al., 2022), and TCN (Mahapatra & Bhuyan, 2022) for richer temporal modelling. Attention mechanisms have further been integrated into CNN pipelines to focus learning on the most task-relevant features (Datta et al., 2021).

Beyond standard classification pipelines, three specialised learning strategies have been applied to imagined speech decoding. Transfer learning has been the most widely adopted, serving four purposes: adapting pretrained architectures such as ResNet, DenseNet, and AlexNet from large datasets (Chinta & Moorthi, 2022; Mahapatra & Bhuyan, 2023; Panachakel &

Ganesan, 2021); reducing inter-individual differences through domain adaptation (Jiménez-Guarneros & Gómez-Gil, 2021) and codebook-based augmentation (García-Salinas et al., 2019); translating imagined to overt speech via CNN encoder-decoder models (D.-H. Lee et al., 2023; Rekrut et al., 2022); and fine-tuning on multi-subject data for target individuals (Niimura et al., 2023). Incremental learning has addressed the vocabulary expansion problem, with García-Salinas et al. (2023) employing k-means centroid-based CNN retraining and Lee et al. (2021) using Siamese networks with deep metric learning for new class detection. Multimodal learning has compensated for EEG's spatial and SNR limitations by fusing EEG with fNIRS (Cooney et al., 2022) and surface electromyography. Additional approaches include Siamese networks (Heting & Nuo, 2022), reinforcement learning (Mahapatra & Bhuyan, 2022), and prototype networks (Hernandez-Galvan et al., 2022). In continuous decoding, only two studies have attempted full sentence reconstruction from imagined speech, using adversarial domain adaptation with HiFiGAN and HuBERT and TCN-based context-aware word embeddings with sliding windows (J.-W. Lee et al., 2023).

Together, these modelling choices, from the selection of shallow versus deep architectures to the use of transfer and adaptation strategies, form the technical substrate on which cross-subject generalisation must ultimately be built, and it is the fragility of that substrate across individuals that the remainder of this review examines. It is important to note that the field faces several documented structural constraints that compound the cross-subject generalisation problem: datasets remain small, fragmented, and non-standardised; preprocessing pipelines are dataset-specific and poorly suited to real-world deployment; and decoding architectures are largely limited to discrete unit classification rather than the continuous sentence-level output that clinical use requires (Jin et al., 2025). A cross-cutting concern is user privacy, as imagined

speech decoding systems inherently access internal linguistic content, including unintended cognitive material beyond the target signal, making privacy a structural rather than peripheral challenge for any deployment framework.

Cross-Subject Generalisation and the Path to Universal Decoding

Of all the challenges facing imagined speech BCI systems, cross-subject generalisation is the one that most directly determines whether the technology can move from a subject-specific laboratory tool to a universally deployable assistive device. EEG signals during imagined speech exhibit pronounced inter-individual variability arising from anatomical differences in cortical folding, individual variation in language network organisation, differences in cognitive strategy during the imagery task, and inconsistencies in electrode placement across recording sessions (Jin et al., 2025). The consequence is a domain shift between the data distribution of training subjects and that of any new user, a shift that standard supervised learning has no mechanism to handle without retraining or adaptation.

The scale of this challenge has been empirically confirmed. Subject-dependent models, trained and tested on data from the same individual, consistently and substantially outperform subject-independent approaches, a gap that has been replicated across datasets, architectures, and feature sets (Liyanagoonawardena et al., 2023). This performance gap is not simply a matter of needing better models; it reflects a fundamental mismatch between how current systems learn and how neural signals vary across people.

Several technical approaches have been brought to bear on this problem. Domain adaptation and transfer learning methods attempt to bridge the distributional gap between source and target subjects, either by aligning feature distributions or by fine-tuning pretrained models on small amounts of target subject data (Peng et al., 2023; Zhao et al., 2024). The limitation here is

practical as much as technical: these methods typically require some data from the new user for adaptation, which reintroduces a calibration burden that undermines the goal of a universally deployable system. Approaches that extract shared underlying features common across subjects aim to sidestep this requirement by learning representations that are invariant to individual differences, though doing so risks discarding the subject-specific signal nuances that contribute to accurate decoding. Meta-learning strategies, designed to produce models that can adapt rapidly to new users from only a small number of calibration samples, represent a more recent and promising direction, though their performance in the imagined speech domain remains to be rigorously established across diverse subject populations (Jin et al., 2025).

The technical questions surrounding cross-subject generalisation do not exist in isolation from broader questions about what kind of technology is being built and for whom. Imagined speech decoding involves the acquisition and processing of neural data that encodes inner linguistic content, data that occupies a uniquely sensitive position in the landscape of personal information and that cannot be straightforwardly anonymised given the identifying properties of individual EEG signatures (Ienca & Haselager, 2016; Kellmeyer, 2018). The populations most consistently invoked as the intended beneficiaries of this technology, those living with ALS, locked-in syndrome, or severe motor impairment, are also among those least positioned to navigate the consent, calibration, and data-sharing demands that current systems impose. These are not concerns that await the technology's maturity before becoming relevant. They are present in the design choices, evaluation standards, and reporting practices of the research being conducted now, and examining them honestly is part of what this review sets out to do.

Research Questions

The primary research question guiding this review is: *What is the current state of cross-subject generalization in EEG-based imagined speech decoding systems, and what technical, methodological, and dataset-related factors influence it?*

A secondary research question aims to address *what governance, ethical, and regulatory implications emerging from the identified generalization challenges*. The primary research question guided all screening, extraction, and synthesis procedures. The secondary research question was addressed interpretively in a dedicated section drawing on the synthesized evidence, and did not influence study eligibility criteria.

Organisation of This Review

This review is organised to move from foundational grounding to empirical synthesis to interpretive argument. Following this introduction, Section 2 details the methodology underpinning the systematic search and screening process, including the eligibility criteria, search strategy, data extraction procedures, and quality assessment framework developed for this review, conducted in accordance with PRISMA 2020 guidelines. Section 3 presents the results of the systematic review across nine included studies, organised around four interconnected themes: the cross-subject performance landscape, the calibration burden, methodological fragmentation, and governance implications. Section 4 draws on the synthesised evidence to develop the central analytical argument of this review, that cross-subject generalisation and calibration burden are not independent technical problems but two sides of a single tradeoff whose implications for clinical deployment and equitable access have not been adequately examined in the literature. Section 5 addresses the governance and ethical dimensions that follow from this analysis, considering what the current state of the evidence implies for regulatory readiness, research reporting standards, and the populations this technology is intended to serve. Section 6 concludes

by identifying the specific conditions under which meaningful progress on the generalisation problem would constitute genuine advancement toward deployment, and what the field would need to establish before that claim could be responsibly made.

Methods

This systematic review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al., 2021).

Search Strategy

The search strategy was developed in four phases: identification of key terms, construction of a Boolean string, adaptation across databases, and application to screening. Three conceptual blocks were used: (1) EEG and related acquisition terms, (2) non-overt speech paradigm terms including imagined speech, inner speech, covert speech, speech imagery, and speech imagination, and (3) machine learning and decoding terms including classification, deep learning, neural networks, CNN, LSTM, and transformer. Cross-subject terms were deliberately excluded from the Boolean string, as the phenomenon of interest is the paradigm itself rather than a specific evaluation approach. The final search string was:

((("imagined speech" OR "inner speech" OR "covert speech" OR "speech imagery" OR "speech imagination" OR "silent speech") AND (EEG OR electroencephalograph)) AND ("machine learning" OR classification OR decoding OR "deep learning" OR "neural network*" OR CNN OR LSTM OR transformer*))*

Searches were conducted in Scopus, Web of Science, PubMed, and IEEE Xplore, covering publications from 2014 to February 2026. Two rounds of search were conducted in January and February 2026, one week apart using the same Boolean search string and procedures

to check the effectiveness of search string as well as ensure consistency within results. Backward and forward citation searches were conducted as a supplementary step using Google Scholar and Scopus in another week of gap.

Study Selection

Retrieved papers were imported into Covidence systematic review software (CovidenceVeritas Health Innovation, 2023) for deduplication and track decisions during title and abstract screening and full time review. During each stage of selection and screening, the papers were also categorically managed in Google Sheets. References were managed using Zotero (Corporation for Digital Scholarship, 2023).

Inclusion and Exclusion Criteria

Studies were eligible if they met all of the following criteria: (1) EEG was the primary signal acquisition modality; (2) the paradigm involved imagined, inner, covert, or silent speech with no audible or visible output; (3) a machine learning or computational decoding approach was employed; (4) the study was an original empirical research article; and (5) the study reported cross-subject or subject-independent evaluation as defined below.

Studies were excluded if any of the following applied: non-EEG or mixed modality signal acquisition; overt speech or speech perception paradigms with no imagined component; absence of a classification or decoding task; non-primary source format such as reviews, editorials, or abstracts without full text; non-English language; non-human participants; or failure to meet the operational definition of cross-subject generalization.

For this review, cross-subject generalization is defined as the ability of a decoding model to classify EEG signals from a subject whose data were entirely absent from the training set (Brookshire et al., 2024). This requires a strict subject-level partition between training and test

data, such that no trial, session, or signal from the test subject contributes to model training, hyperparameter selection, or feature extraction (Becker & Recamonde-Mendoza, 2026; Del Pup et al., 2025; Ding et al., 2026; Rosenblatt et al., 2024; Shim et al., 2021). Studies that trained and evaluated models on pooled or randomly shuffled multi-subject datasets without enforcing subject-level holdout were not considered to meet this criterion, regardless of the terminology used by the original authors.

Screening Procedure

Screening included removing duplicates, reviewing abbreviated information, and conducting a full-text review. Covidence was used to identify the duplicates and were manually checked for. Next, each article's title, keywords, and abstract were screened and the ones that met the exclusion criteria cleanly were removed first including non-primary sources like book chapters, literature reviews, and meta analyses. Papers with sufficient ambiguity were advanced to the next stage. For the full-text stage, articles were read in the following order: Abstract, Methods, Results, Discussion and Introduction. If an article did not meet the exclusion criteria, it was read twice to gauge whether it fit my research questions. During the second full-text review, references cited in the included studies were identified and then considered based on their titles, and contextual descriptions. Evaluation for these referenced articles, using the same strategy of an abbreviated screening, and then a full-text review.

Data Items and Extraction

Data extraction was performed by a single reviewer using a structured form applied consistently across all nine included studies. Variables were organised according to the two research questions.

Primary Research Question

For each study, the following were recorded: *publication details* (year, venue); *dataset characteristics* (number of subjects, EEG channels, sampling rate, speech tasks, and public availability); *covert speech paradigm type*; *feature extraction methods and model architecture*, distinguishing between traditional machine learning, deep learning, and hybrid approaches.

The *cross-subject generalization strategy* was extracted in detail, including the specific evaluation protocol used. Where subject-level partitioning was explicitly stated, this was recorded directly; where it required inference from the described methodology, this was noted; and where it could not be verified, this was flagged as a study-level limitation. The *amount of target subject data required for adaptation or fine-tuning* was also extracted where reported, to capture the calibration burden associated with each approach. *Reported performance metrics* were recorded alongside the presence or absence of statistical validation.

Secondary Research Question

For each study, concerns, if any, related to *privacy, data sensitivity, misuse, or equitable access* were acknowledged. The absence of such discussion was recorded as meaningfully as its presence, given its relevance to the governance analysis.

Quality Assessment

Given the absence of an established quality assessment tool applicable to technical machine learning studies of this nature, a custom evaluation framework was developed. This approach is consistent with systematic review practice in emerging technical fields where standardised instruments designed for clinical or interventional research do not map meaningfully onto the methodological concerns of the domain under review. Each included study was assessed narratively across the following dimensions, with patterns noted across the full set of included studies rather than scores assigned to individual papers.

1. *Subject-level partitioning clarity* refers to whether the cross-subject evaluation protocol was explicitly described in sufficient detail to verify that no data from the test subject contributed to model training, hyperparameter selection, or feature extraction. This dimension is central to the primary research question and directly reflects the operational definition applied during screening.
2. *Sample size adequacy* considers whether the number of subjects included in cross-subject evaluation was sufficient to draw meaningful conclusions about generalizability, noting that very small subject pools limit the interpretability of reported performance.
3. *Statistical validation* captures whether reported performance metrics were accompanied by appropriate statistical treatment, such as significance testing, confidence intervals, or cross-validation procedures that adequately guard against overfitting.
4. *Dataset independence and transparency* assesses whether the dataset used was publicly available, whether its characteristics were sufficiently described, and whether the same dataset appears across multiple included studies in ways that may limit the independence of findings.
5. *Calibration burden transparency* refers to whether studies clearly reported the amount of target subject data required for model adaptation or fine-tuning, which is directly relevant to the practical deployability of the proposed approach.
6. *Methodological reproducibility* considers whether the described methods were reported in sufficient detail to permit independent replication.

Weaknesses identified through this assessment are interpreted in context throughout the synthesis. Where limitations are consistent across multiple included studies, these are treated as

field-level observations rather than criticisms of individual works, and are discussed accordingly in the governance implications section.

Synthesis Approach

Given the heterogeneity of methods, datasets, and evaluation protocols across included studies, a narrative synthesis was conducted which was organised thematically, with themes derived from the primary and secondary research questions.

Four interconnected themes structure the synthesis. The first, *the cross-subject performance landscape*, characterises what verified cross-subject decoding accuracy looks like across the included studies once methodologically inflated results are set aside. This theme examines reported performance in relation to evaluation protocol, dataset used, and task complexity, with particular attention to whether claimed results hold under strict subject-level partitioning.

The second theme, *the calibration burden*, examines the extent to which cross-subject generalization in included studies is conditional on access to target subject data for model adaptation or fine-tuning. Rather than treating calibration as a technical detail, this theme treats it as a substantive dimension of generalizability, asking what each approach actually requires from a new user and what that implies for populations who cannot easily provide it.

The third theme, *methodological fragmentation*, documents the inconsistency of evaluation standards, terminology, and dataset usage across the included literature. This theme considers not only individual study limitations but the cumulative effect of fragmented methodology on the field's ability to make genuine cumulative progress on the generalization problem.

The fourth theme, *governance implications*, draws interpretively on the three preceding themes to address the secondary research question. Rather than treating governance as a separate discussion appended to technical findings, this theme treats the identified technical and methodological limitations as the empirical basis for a governance argument: that a technology whose generalization is calibration-dependent, whose performance is systematically difficult to verify, and whose evaluation standards remain inconsistent, poses specific and underappreciated challenges for regulatory frameworks, clinical translation, and equitable access.

Together these four themes are intended to advance beyond a descriptive catalogue of existing methods toward a coherent analytical account of where the field stands, why progress has been slower than reported accuracies might suggest, and what governance structures are needed before imagined speech BCIs move closer to real-world deployment. This organising framework, which this review terms the *generalization-calibration tradeoff*, has not to the authors' knowledge been explicitly proposed as a unifying lens in the imagined speech BCI literature, though its constituent observations are grounded in and traceable to the extracted evidence.

Results

Prisma flowchart for results depicted in Figure 1.

Screening Procedure

Database searches retrieved 720 records in total (Scopus: 104, Web of Science: 153, PubMed: 91, IEEE Xplore: 372). After deduplication, 311 unique records remained.

Title and abstract screening was conducted by a single reviewer. As a preliminary step, 185 records were examined to assess whether papers primarily concerned with cross-subject generalization consistently signal this in title, abstract, or keywords. This examination confirmed

that all such papers used identifiable terms including cross-subject, subject-independent, inter-subject, domain adaptation, transfer learning, meta-learning, or generalization. On this empirical basis, records with no such signal were excluded at the title and abstract stage as unlikely to address the primary research question. Ambiguous cases were advanced to full-text review. This procedure is acknowledged as a potential source of exclusion bias and is documented as a limitation.

Of 311 records, 276 were excluded at title and abstract stage. The majority of exclusions were attributable to clear and verifiable criteria: non-primary source format (32); wrong or mixed signal modality (31); wrong speech task (29); no machine learning or decoding component (5); and other reasons including non-English language, animal studies, and hardware-focused papers (28). The largest exclusion category comprised 151 records with no cross-subject signal in title, abstract, or keywords. This is consistent with the composition of the broader literature: within this group, a substantial proportion were either explicitly subject-dependent by stated design or inferably so from the described methodology or framing, with the remainder simply not addressing generalization as a research concern in any form. Separately, 12 records from the full screened pool carried sufficient ambiguity regarding their evaluation approach to warrant advancement to full-text review despite the absence of clear cross-subject framing.

To assess the validity of this exclusion decision, a random sample of 25 records was drawn from the 151 excluded on the basis of absent cross-subject signal and retrieved for full-text review. Of these, 23 were confirmed as subject-dependent by explicit statement of design, one did not involve a classification or decoding task, and one was not primary empirical research articles. None were found to meet the operational definition of cross-subject generalisation applied in this review. This post-hoc verification provides empirical support for

the keyword-based exclusion heuristic and reduces, though does not fully eliminate, the risk that eligible studies were excluded at this stage.

The remaining 35 records advanced to full-text screening. Of these, 26 were excluded at the full-text stage. Exclusion reasons were as follows: subject-dependent design confirmed at full-text (10); different speech task (2); and methodological concerns precluding verification of subject-level holdout, primarily pooled multi-subject designs presented as subject-independent without enforcing trial-level separation (14). A consistent pattern was observed across the latter group: studies claiming cross-subject or subject-independent evaluation had pooled or randomly shuffled data across subjects without a strict subject-level partition, a design that did not meet this review's operational definition. This pattern is discussed as a field-wide methodological inconsistency in the synthesis.

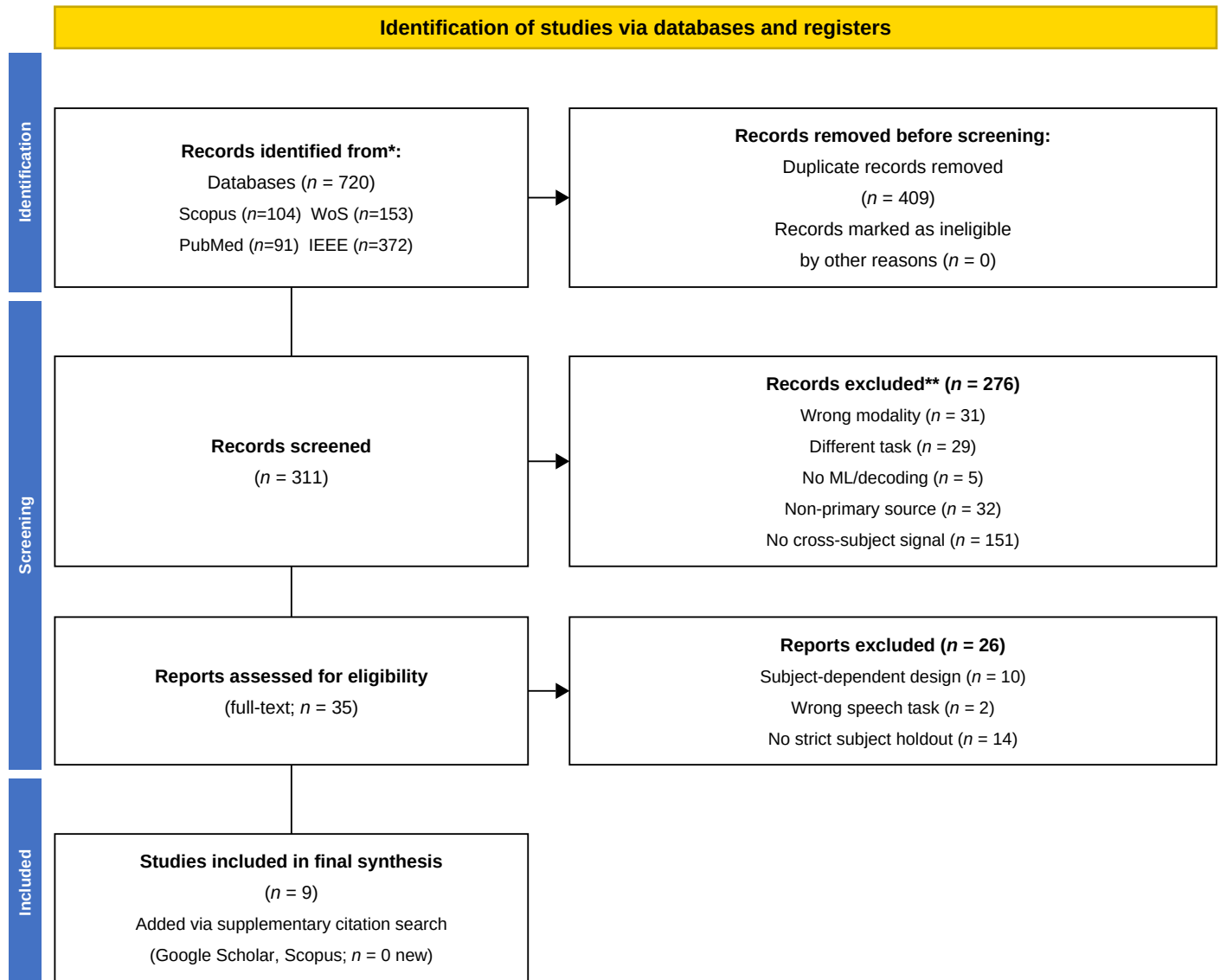
Nine studies were confirmed as meeting all eligibility criteria and were included in the final synthesis. Single-reviewer screening is acknowledged as a limitation consistent with the thesis context.

Overview of Included Studies

The systematic search and screening process yielded nine studies meeting all eligibility criteria. The nine studies included in this review were published between 2019 and 2025, with two appearing in 2019, one in 2021, one in 2023, three in 2024, and two in 2025. Publication venues span peer-reviewed journals and conference proceedings across signal processing, pattern recognition, and brain-computer interface research, including Pattern Recognition Letters, IEEE Transactions on Human-Machine Systems, Elsevier journals, and conference proceedings at ICASSP, Interspeech, CAIP, and APSIPA ASC.

Figure 1

PRISMA Flow Diagram for New Systematic Review, Which Included Searches of Databases, Registers and Other Sources Demonstrating Search Strategy



Note. This flow diagram illustrates the results of the search and selection process, from the number of records identified in the search to the number of studies included in the systematic review.

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. doi: 10.1136/bmj.n71

Across the nine studies, six distinct datasets were used. The Kara One dataset, released by the University of Toronto in 2015, appears in three studies - Saha and Fels (2019a), Saha, Abdul-Mageed and Fels (2019b), and Zhang, Ding and Chen (2025) - making it one of two most frequently used dataset in the included set. The Thinking Out Loud dataset (Nieto et al., 2022) also appears in three studies - Radwan et al. (2024), Ng and Guan (2023), and Ng and Guan (2024). The remaining studies draw on a variety of sources: Jiménez-Guarneros and Gómez-Gil (2021) use two separate datasets, a 15-subject Spanish words dataset and a 6-subject long words dataset; Bhalerao and Pachori (2025) test across three datasets - the Coretto dataset (15 subjects), the BCI Competition Track 3 dataset (15 subjects), and a Motor and Speech Imagery EEG dataset from the University of Malta (5 subjects); and Guo et al. (2024) use a proprietary dataset collected by the authors, the only study in the included set to do so. No dataset outside the Kara One and Thinking Out Loud collections appears in more than one study.

Subject counts across the included studies range from five to 54. The majority of studies involved fewer than 20 subjects: the Kara One studies worked with 14 subjects in two cases and eight subjects (after quality filtering) in one; the Thinking Out Loud studies each involve 10 subjects; the Spanish words dataset contributes 15 subjects, the long words dataset six, the Coretto and BCI Competition datasets 15 each, and the Malta dataset five. Guo et al. (2024) report the largest subject pool at 54, all of whom are described as right-handed healthy adult males. Across all nine studies, none reported recruiting participants with ALS, locked-in syndrome, or any other motor or speech impairment, despite clinical populations of this kind being explicitly invoked as the intended beneficiaries of the technology in the introductions of seven of the nine papers.

The nine studies represent a wide range of decoding approaches. Saha and Fels (2019a) and Saha, Abdul-Mageed and Fels (2019b) both employed a hybrid architecture combining spatial and temporal CNNs with a deep autoencoder and XGBoost classifier, applied to cross-covariance matrix features. Jiménez-Guarneros and Gómez-Gil (2021) used a bidirectional GRU with unsupervised domain adaptation via adaptive batch normalisation and a variation of information loss. Ng and Guan (2023) proposed a Siamese variational autoencoder paired with DeepConvNet or EEGNet classifiers. Ng and Guan (2024) applied a MAML-inspired meta-learning framework with DeepConvNet as the base learner. Guo et al. (2024) used a temporal-spatial CNN encoder pretrained on overt speech EEG with mel-spectrogram supervision and fine-tuned on covert EEG. Zhang, Ding and Chen (2025) proposed a multi-band CNN encoder with gated band fusion, bidirectional LSTM, and a prototypical classification head using a hybrid distance metric. Radwan et al. (2024) compared over twenty machine learning and deep learning models including a proposed BruteExtraTree classifier alongside ResNet18, GRU, EEGNet, and a convolutional VAE. Bhalerao and Pachori (2025) used a multivariate signal decomposition method with pretrained CNN feature extractors and a quadratic SVM classifier.

Leave-one-subject-out cross-validation is the stated or implied evaluation framework in seven of the nine studies, though the degree to which subject-level partitioning is explicitly verified varies considerably across the set. Jiménez-Guarneros and Gómez-Gil (2021) described LOSOCV in full procedural detail and repeated it five times with different partitions and random initialisations. Ng and Guan (2024) explicitly described a subject-disjoint train, validation, and test partition with paired significance testing. Guo et al. (2024) applied LOSO for pretraining combined with 5-fold cross-validation on the target subject's data. Saha and Fels (2019a) used LOSO in Phase Two but pooled random splitting in Phase One for hyperparameter selection.

Saha, Abdul-Mageed and Fels (2019b) reported headline results from a pooled random split and used LOSO selectively for comparison tables and the confusion matrix, without reporting a complete LOSO accuracy for the primary 11-class task. Zhang, Ding and Chen (2025) stated LOSO but required 33 labeled samples from the held-out subject to construct class prototypes at test time. Ng and Guan (2023) implied subject-level holdout without formally describing the partitioning procedure. Bhalerao and Pachori (2025) described 10-fold cross-validation and leave-one-out cross-validation in terms that are not fully reconcilable, without confirming that test subjects were entirely absent from training folds. Radwan et al. (2024) used a single stratified train-test split with no cross-validation.

The amount of target subject data required at adaptation or inference time varies substantially across the nine studies. Saha and Fels (2019a) and Saha, Abdul-Mageed and Fels (2019b) required no target subject data in any form. Bhalerao and Pachori (2025) likewise reported no target subject fine-tuning requirement. Jiménez-Guarneros and Gómez-Gil (2021) required no labeled target data but used 80% of the target subject's unlabeled EEG during the adaptation phase. Ng and Guan (2024) tested both a zero-shot condition and a few-shot condition requiring 50 labeled trials from the target subject, reporting performance under both. Ng and Guan (2023) required 50 labeled trials from the target subject for the subject-adaptive step, which produces the best reported results. Zhang, Ding and Chen (2025) required 33 labeled trials from the held-out subject at inference time to construct class prototypes, without which classification is not possible. Guo et al. (2024) used 80% of the target subject's labeled covert EEG for fine-tuning, with zero-shot performance separately reported at 24.5% on a 5-class task. Radwan et al. (2024) did not specify the calibration requirement for the subject-independent condition.

Reported performance metrics are not uniform across the set. Accuracy is reported in all nine studies. Cohen's Kappa is additionally reported in Saha and Fels (2019a), Saha et al (2019b), and Zhang et al (2025). F1 score is reported in Guo et al. (2024) and Zhang et al (2025). Precision and recall are reported in Zhang et al (2025) and Saha et al (2019b). Standard deviations accompany accuracy figures in six of the nine studies; they are absent in Saha et al (2019a), Saha et al (2019b), and Radwan et al. (2024). Statistical significance testing is present in three studies - Jiménez-Guarneros and Gómez-Gil (2021) applied Wilcoxon signed-rank and Friedman tests with Finner correction; Ng and Guan (2024) reported paired t-tests with p-values; and Guo et al. (2024) reported a p-value for the pretraining ablation comparison. The remaining six studies report no formal statistical testing of their cross-subject results.

None of the nine included studies contains a substantive discussion of privacy, data sensitivity, consent, or equitable access in relation to their proposed systems. Eight studies make no mention of these dimensions at any point. Ng and Guan (2024) briefly acknowledged that long calibration periods impose a burden on users with disabilities, without further elaboration. Clinical populations, including individuals with ALS, locked-in syndrome, and motor impairment, are referenced in the introductions of seven studies as the intended beneficiaries of imagined speech BCI technology, but no study describes recruitment of such populations, discusses deployment constraints specific to them, or addresses the governance implications of systems that process inner linguistic content.

Full study characteristics are presented in Table 1 and evaluation details in Table 2.

Descriptive Summary

Radwan et al. (2024) offered the broadest model comparison in the included set, evaluating over twenty machine learning and deep learning classifiers on the Thinking Out Loud

Table 1*Methodological Characteristics of Included Studies*

Author(s) & Year	Dataset (Name, Subjects, Channels, SR)	Study Objective	Paradigm Type	Feature Extraction	Model / Architecture
Jimenez-Guarneros & Gomez-Gil (2021)	Long words: 6 subj, Spanish words: 15 subj, 14 ch, 128 Hz; calibration: 64 ch, 1000 Hz	Propose SRDA, an unsupervised domain adaptation method combining AdaBN and VOI loss for calibration-free cross-subject imagined speech classification	Imagined speech	Clustering-based codebook histogram (k-means, k=500/650)	Bi-directional RNN (SRDA)
Ng & Guan (2023)	Thinking Out Loud: 10 subj, 128 ch, 1024 Hz (↓ to 254 Hz)	Propose a Siamese spatiotemporal encoder with contrastive loss for unsupervised domain generalization of inner speech EEG	Inner speech	Spatiotemporal VAE with selective contrastive loss for selective data rejection	Siamese Spatiotemporal VAE / EEGNet
Guo et al. (2024)	Authors' own dataset: 54 subj, 64 ch, 5000 Hz (↓ to 400 Hz)	Propose a two-stage knowledge transfer framework using overt EEG pretraining with audio supervision followed by covert fine-tuning for cross-subject decoding	Covert speech	Spatial-temporal conv with mel-spectrogram audio supervision (80 mel-filters)	CNN two-stage (LG framework)
Zhang et al. (2025)	Kara One: 8 of 14 subj, 62 ch, 256 Hz	Propose TeleProNet, a meta-learning prototypical network with multiband feature extraction for few-shot subject-independent imagined speech classification	Imagined speech	Multi-band envelope via FIR filter and Hilbert transform (Mu, Beta, Gamma, Delta)	TeleProNet (gated CNN + Prototypical Net)
Saha, Fels & Abdul-Mageed (2019)	Kara One: 14 subj, ch count not reported, SR not reported	Develop a subject-independent hierarchical CNN-LSTM-DAE and XGBoost framework for phonological category classification from imagined speech EEG	Imagined speech	Channel cross-covariance (CCV) matrices across all electrode pairs	Hierarchical DAE (CNN-LSTM-DAE + XGBoost)
Radwan et al. (2024)	Thinking Out Loud: 10 subj, 128 ch, 1024 Hz (↓ to 254 Hz)	Benchmark ML and DL models in subject-dependent and subject-independent settings; propose BruteExtraTree as a novel solution to overfitting	Inner speech	PCA, ICA, DWT, CSP (pairwise and one-vs-all), CAR+DWT, and concatenation of all	BruteExtraTree, ConvVAE + FCN, DeepConvNet (benchmark)
Bhalerao & Pachori (2025)	Coretto: 15 subj, 6 ch; BCI Competition: 15 subj, 64 ch; Univ. of Malta: 5 subj, 32 ch	Propose an MSSDM-based framework for cross-subject and cross-dataset imagined speech classification using multivariate joint time-frequency features	Imagined speech	MSSDM: GA-optimized swarm filter banks + sparse Fourier transform + Hilbert transform to produce multivariate JTF images	MSSDM (quadrature feature-based classifier)
Saha, Abdul- Mageed & Fels (2019)	Kara One: 14 subj, 64 ch, SR not reported	Propose a two-phase hierarchical deep learning framework for subject-independent phonological category classification (Phase 1) and individual speech token identification (Phase 2)	Imagined speech	CCV matrices for Phase 1; stacked DAE latent vectors from 6 phonological classifiers for Phase 2	Two-Phase DAE (Phase 1 + 2)
Ng & Guan (2024)	Thinking Out Loud: 10 subj, 128 ch, 1024 Hz (↓ to 254 Hz); also evaluated on motor imagery datasets	Propose a MAML-inspired meta-learning framework treating each subject as an independent task for rapid generalization to unseen subjects with zero or minimal target data	Inner speech	Raw EEG processed end-to-end by DeepConvNet; no handcrafted feature extraction	DeepConvNet (meta-learning adaptive)

Note. Studies are ordered chronologically within each paradigm type. SR = sampling rate; ch = channels; subj = subjects; LOSO = leave-one-subject-out cross-validation; DAE = denoising autoencoder; CNN = convolutional neural network; VAE = variational autoencoder; MAML = model-agnostic meta-learning.

Table 2*Evaluation and Performance Summary of Included Studies*

Author(s) & Year	Cross-Subject Generalization Strategy	Evaluation Protocol	Calibration Burden	Cross-Subject Performance & Metric
Jimenez-Guarneros & Gomez-Gil (2021)	Unsupervised domain adaptation (SRDA): AdaBN for marginal alignment + VOI loss for conditional alignment; no target labels required	LOSO CV repeated 5× with different partitions; 80% source + target for adaptation, 20% for testing	Zero-shot; no labeled target data required at any stage	61.02 ± 8.14% accuracy (5-class, Spanish words); 62.99 ± 4.78% (2-class, Long words)
Ng & Guan (2023)	Domain generalization via VAE lossy compression to suppress subject-specific features; optional adaptive transfer learning with selective contrastive learning	LOSO; fine-tuning on 50 labeled target trials; zero-shot and ablation across framework configurations	Low; base framework is zero-shot; adaptive step requires 50 labeled target trials	29.92% mean accuracy (proposed + DeepConvNet); baselines: 24.54% (EEGNet), 26.68% (DeepConvNet); chance = 25%
Guo et al. (2024)	LOSO overt pretraining on 53 subjects to build base model; subject-adaptive covert fine-tuning on 80% of target subject trials; overt-to-covert knowledge transfer via audio supervision	LOSO pretraining; 5-fold CV on target subject covert data (4 folds fine-tune, 1 fold test); zero-shot (0%) also tested	Moderate; ~80 target subject covert trials required for fine-tuning; zero-shot tested at 24.5 ± 6.0%	32.2 ± 9.1% accuracy; F1: 29.7 ± 9.3% (5-class, subject-adaptive); zero-shot: 24.5 ± 6.0%
Zhang et al. (2025)	Meta-learning with prototypical networks; episodic training with support sets drawn from 3 different training subjects per episode to capture cross-subject diversity	LOSO over 8 subjects; 100 test episodes per fold; support set (3 samples/class) and query set (4 samples/class) drawn exclusively from held-out subject	Minimal; only 3 labeled support samples per class (33 total) from target subject required at test time; no target data used in training	93.84% ± 6.03% accuracy; Precision: 94.05%; F1: 93.82%; Kappa: 0.9322 (11-class, LOSO)
Saha, Fels & Abdul-Mageed (2019)	Subject-independent LOSO classification; no domain adaptation or fine-tuning; fully zero-calibration inference	Phase 1: random 80/10/10% stratified split across all subjects for architecture selection; Phase 2: 14-fold LOSO for fair comparison with prior work	Zero-shot; no labeled target data required at any stage	LOSO avg accuracy: 77.9% (5 binary tasks); range: 73.45%–85.23%; Kappa range: 49.14%–75.92%
Radwan et al. (2024)	Subject-independent pooled train and test split confirmed; no subject overlap between train and test; no domain adaptation or fine-tuning applied	Single stratified split across all 10 subjects with no subject overlap; no explicit LOSO or k-fold structure for subject-independent case	Zero-shot; no labeled target data used in subject-independent setting	BruteExtraTree + PCA: 32%; ConvVAE + FCN: 32%; ShallowFBCSPNet: 29%; chance = 25% (4-class)
Bhalerao & Pachori (2025)	LOSO with 10-fold CV across all subjects; cross-dataset validation: trained on one dataset, tested on another using common commands; no fine-tuning applied	10-fold CV across all subjects within each dataset; cross-dataset validation between Coretto and Univ. of Malta; average accuracy and SD reported per subject and overall	Zero-shot; no labeled target data required for cross-subject or cross-dataset evaluation	Coretto (6-class): 60.80 ± 1.13%; BCI Competition (5-class): 59.10 ± 1.68%; cross-dataset results also reported
Saha, Abdul- Mageed & Fels (2019)	Subject-independent classification; primary evaluation: stratified split across all subjects; LOSO also performed for comparison with prior work	Primary: 80/10/10% stratified split; secondary: 14-fold LOSO; multiple experiments conducted for train/test ratio robustness	Zero-shot; no labeled target data required at any stage	Phase 1 (80/10/10): 84.27% avg across 6 binary tasks (range: 78.33%–89.16%); Phase 2: 53.36% (phonological features) vs. 28.08% (direct CCV); LOSO avg: 77.9%
Ng & Guan (2024)	MAML-inspired meta-learning; each subject treated as an independent task; meta-loss encourages rapid generalization; optional few-shot subject-adaptive fine-tuning	LOSO (8 train, 1 validation, 1 test); zero-shot and few-shot (50 target trials) conditions compared against baselines including standard transfer learning and EEGNet	Zero to low; zero-shot: 25.55 ± 2.17%; few-shot (50 trials): 31.15 ± 1.82%; substantially lower than prior transfer learning methods requiring 160+ trials	Zero-shot: 25.55 ± 2.17%; few-shot (50 trials): 31.15 ± 1.82%; median: 30.79%; chance = 25%

Note. Cross-subject performance metrics are reported as mean ± standard deviation where available; chance-level accuracy is 25% for 4-class tasks and varies by study. LOSO = leave-one-subject-out cross-validation; CV = cross-validation; F1 = F1-score; SD = standard deviation.

dataset across both subject-dependent and subject-independent conditions. The dataset comprises 10 subjects performing four-class inner speech tasks consisting of directional Spanish words, yielding 2,236 trials in total. Feature extraction explored multiple strategies including ICA, PCA, DWT, CSP in pairwise and one-versus-all configurations, and raw data, with no single preprocessing approach dominating across classifier families. The proposed BruteExtraTree classifier, a stochastic ensemble method that trains between one thousand and 2,500 decision trees with varying random seeds and selects the best-performing seed on a development set, matched the best deep learning result, a convolutional VAE with fully connected head, at 32% accuracy in the subject-independent condition, against a chance level of 25%. Subject-dependent performance reached 46.6% for the same classifier. The subject-independent evaluation partitioned subjects between training and test sets, though no cross-validation or repeated splitting was applied, and the single stratified split was not accompanied by any statistical significance testing. The authors note explicitly that the 32% figure is only marginally above chance and attribute this to noise levels and recording inconsistencies rather than classifier limitations.

Bhalerao and Pachori (2025) proposed a joint time-frequency framework grounded in Multivariate Signal Subspace Decomposition Method, which decomposes EEG into channel-aligned oscillatory components and applies a Hilbert transform to generate multivariate joint time-frequency images. These images are passed through five pretrained neural network architectures, ELM, LeNet, AlexNet, SqueezeNet, and ResNet, to extract deep features, which are then reduced via either canonical correlation analysis or hyper-dimensional computing before final classification with a quadratic SVM. The study evaluates across three public datasets: the Coretto dataset with 15 subjects and six speech classes, the BCI Competition Track 3 dataset

with 15 subjects and five classes, and a Motor and Speech Imagery EEG dataset from the University of Malta with five subjects and four classes, all downsampled to 1,024 Hz. The best within-dataset result, 60.80% plus or minus 1.13% on the six-class Coretto task, was achieved by the ResNet feature extractor combined with canonical correlation analysis reduction.

Cross-dataset experiments trained on one dataset and tested on another, with results varying substantially across model and dataset combinations. The paper additionally reports joint time-frequency visualisations showing similar oscillatory patterns across subjects in the theta and alpha bands during word imagination. The evaluation protocol describes 10-fold cross-validation with a 70/30 split per fold, though the paper also references leave-one-out cross-validation without reconciling the two procedures, and does not explicitly confirm that test subjects were withheld from training folds in the within-dataset experiments.

Saha and Fels (2019) framed imagined speech decoding as an explicitly subject-independent problem, targeting five binary phonological classification tasks on the Kara One dataset across 14 subjects using 62 EEG channels. Feature extraction relies on channel cross-covariance matrices, which capture inter-electrode joint variability across all channel pairs rather than treating channels independently. These matrices are passed through a hierarchical hybrid architecture consisting of a two-dimensional spatial CNN and an LSTM processing in parallel, whose outputs are concatenated and fed into a deep autoencoder followed by an XGBoost classifier. The study proceeds in two phases: Phase One uses a pooled random 80/10/10 split across all subjects for hyperparameter selection and architecture tuning, reporting average accuracy of 77.9% across the five tasks. Phase Two applies leave-one-subject-out cross-validation across all 14 subjects, with the reported standard deviation across subjects collapsing from approximately 22% in prior baseline methods to 5.41% under the proposed

model. LOSO performance ranged across approximately 73 to 85% depending on the phonological task, with the consonant-versus-vowel distinction producing the clearest subject-level clustering in t-SNE visualisations of the cross-covariance feature space, while the nasal-versus-non-nasal task produced the most diffuse cluster structure. No calibration data from the test subject was used at any stage of Phase Two evaluation.

Saha, Abdul-Mageed and Fels (2019) extended the same laboratory's cross-covariance and deep autoencoder framework into a hierarchical two-phase decoding system on the Kara One dataset, expanding the target vocabulary to 11 speech tokens comprising seven phonemic and syllabic prompts and four words, evaluated across 14 subjects with 64 EEG channels. A sixth binary phonological task distinguishing voiced from voiceless speech is added to the five tasks in the prior paper, and the temporal processing block is replaced with a temporally dilated causal convolution network in place of the LSTM, enabling the model to capture longer-range dependencies without information leakage across time steps. Phase One classifies phonological categories using the same CCV-based architecture, and the latent representations from the deep autoencoder are stacked to form an intermediate feature set for Phase Two, which performs 11-class speech token identification using these phonological predictions as inputs. Primary results for Phase Two reported 53.36% accuracy on an 11-class task using a pooled random split across subjects. Leave-one-subject-out cross-validation is applied selectively in this paper: it appears in the phonological comparison tables and in the confusion matrix for the 11-class task, which under LOSO shows a shift from near-random scattering under raw covariance features to a recognisably diagonal structure when phonological intermediate representations are used. A complete LOSO accuracy for the primary 11-class result is not reported as a single figure. No target subject data is required for either evaluation phase.

Jiménez-Guarneros and Gómez-Gil (2021) proposed a standardisation-refinement domain adaptation method designed to align both marginal and conditional feature distributions between source and target subjects without requiring any labeled data from the target subject. EEG instances from each subject are first encoded as codebook-based occurrence histograms, clustering trials into 500 or 650 codewords depending on the dataset, which produces a fixed-length distributional representation of each trial. A bidirectional GRU serves as the classification backbone across all compared methods. The adaptation procedure combines adaptive batch normalisation for marginal distribution alignment with a variation of information loss for conditional distribution refinement, and exponential moving average for parameter stabilisation. Two datasets are used: a 15-subject Spanish words dataset recorded at 128 Hz with 14 EMOTIV EPOC channels targeting five classes, and a six-subject long words dataset recorded at 1,000 Hz with 64 channels downsampled to 256 Hz targeting two classes. Leave-one-subject-out cross-validation is repeated five times with different partitions and random initialisations, and statistical significance is assessed using Wilcoxon signed-rank and Friedman tests with Finner correction. On the Spanish words dataset, the proposed method achieves 61.02% plus or minus 8.14% against a binomial chance level of 31.42%, a result confirmed as statistically significant. On the long words dataset, aggregate accuracy reaches 62.99% plus or minus 4.78% against a chance threshold of 62.50%, with significance confirmed for four of the six subjects. Although no labeled target data is required, the adaptation phase uses 80% of the target subject's unlabeled EEG recordings, with the remaining 20% used for testing. An ablation study confirms incremental contributions of each adaptation component, and the authors explicitly note that residual negative transfer from source subjects to the target remains an unresolved limitation of the approach.

Ng and Guan (2023) proposed a Siamese variational autoencoder framework for unsupervised domain generalisation in inner speech decoding, targeting the Thinking Out Loud dataset with 10 subjects and four directional speech classes. The VAE encoder follows a spatiotemporal architecture inspired by DeepConvNet, applying spatial filters followed by temporal filters to compress each trial into a 16-dimensional latent representation. A contrastive loss mechanism is used during training to selectively reject training samples whose latent representations are dissimilar to those of the target domain, functioning as an implicit data filtering step before downstream classification. The generalised features are passed to either a DeepConvNet or EEGNet classifier. In addition to the subject-independent baseline, a subject-adaptive variant fine-tunes the downstream classifier on 50 labeled trials from the target subject, which produces the best reported results. The DeepConvNet-based proposed framework achieves 29.92% accuracy against a 25% chance level, compared to a 26.68% subject-independent baseline using EEGNet alone. An ablation confirms that the domain generalisation component contributes the largest single improvement in the DeepConvNet configuration, though in certain EEGNet configurations performance without domain generalisation exceeds the adapted variant. All 10 subjects are reported above chance. Subject-level holdout is implied by the evaluation setup but the partitioning procedure is not formally described, no leave-one-subject-out procedure is stated, and no statistical significance testing accompanies the reported results.

Guo et al. (2024) proposed a two-stage knowledge transfer framework that first pretrains a temporal-spatial CNN encoder on overt speech EEG using concurrent audio recordings as auxiliary supervision, then fine-tunes the pretrained model on the target subject's covert EEG. The dataset is proprietary, comprising 54 right-handed healthy adult males recorded with 64

electrodes at 5,000 Hz downsampled to 400 Hz, performing 100 overt and 100 covert trials per subject across five command phrase classes. During pretraining, the encoder is trained under leave-one-subject-out cross-validation on the 53 remaining subjects' overt EEG, with a root mean square error loss enforcing alignment between the encoder's output and the mel-spectrogram of the simultaneously recorded audio, in addition to a cross-entropy classification loss. Fine-tuning uses 5-fold cross-validation on the target subject's covert data, with four folds used for adaptation and one for testing. The proposed method achieves 32.2% plus or minus 9.1% accuracy and 29.7% plus or minus 9.3% F1 on the five-class task, against a 20% chance level, outperforming the strongest baseline, EEG-Conformer at 27.9% plus or minus 6.5%, by 3.6 percentage points. A zero-shot condition evaluated without any target subject fine-tuning yields 24.5% accuracy, and a few-shot analysis varying the proportion of fine-tuning data from 20% to 80% shows monotonically increasing performance with each increment. The audio supervision component contributes approximately 2.5 percentage points relative to training without it. A statistical significance test is reported for the pretraining ablation comparison.

Ng and Guan (2024) presented a meta-learning framework for subject-independent inner speech decoding, adapting the Model-Agnostic Meta-Learning formulation so that each training subject is treated as an independent few-shot task, explicitly optimising the model for fast adaptation to unseen subjects. DeepConvNet serves as the base learner and receives raw EEG without hand-engineered feature extraction. At each training epoch, a standard cross-entropy classification loss and a meta-learning loss are applied in separate backpropagation steps, with the latter updating parameters to minimise expected loss across all subject-as-task proxies simultaneously. The primary evaluation uses the Thinking Out Loud dataset with 10 subjects, partitioned into eight training subjects, one validation subject, and one test subject in a

leave-one-subject-out scheme, with training and validation subjects playing the role of episode tasks and the test subject remaining entirely disjoint throughout. Two evaluation conditions are reported: a zero-shot condition using no target subject data and a few-shot condition supplying 50 labeled trials from the target subject for fine-tuning. Zero-shot accuracy on the four-class inner speech task is 25.55%, marginally above the 25% chance level, while few-shot accuracy reaches 31.15% plus or minus 1.82%, the highest reported result on this dataset. The paper additionally evaluates on two motor imagery datasets, achieving 87.20% zero-shot and 88.70% few-shot on a 54-subject binary task. No subject across the inner speech evaluation falls below chance level under the few-shot condition. Paired t-tests with reported p-values accompany the primary comparisons.

Zhang et al (2025) proposed TeleProNet, a prototypical network with multi-band feature extraction targeting 11-class imagined speech classification on the Kara One dataset using eight subjects retained after quality filtering. Band-specific features are extracted by applying FIR filters to isolate Mu, Beta, Gamma, and Delta frequency bands, followed by Hilbert envelope computation, channel attention via squeeze-and-excitation modules, and multi-scale temporal convolution with dilation rates of one, two, and three. A gated band-fusion module combines the four band-specific representations, and a bidirectional LSTM with layer normalisation and global average pooling produces the final trial-level embedding. Classification uses a prototypical head with a hybrid Euclidean-cosine distance metric. The model is trained using episodic learning with 11-way 3-shot 4-query episodes, drawing support and query samples from multiple training subjects to promote cross-subject diversity. Evaluation follows a leave-one-subject-out scheme across eight subjects, with 100 test episodes per fold. During inference, three labeled samples per class from the held-out subject are used to construct class prototypes, totaling 33 labeled trials

from the test subject at inference time. The model reports 93.84% plus or minus 6.03% accuracy on the 11-class task, with a Kappa of 0.9322 and a lowest individual subject accuracy of 82.95%. An ablation confirms that the Delta band contributes the largest single-band improvement at approximately 19 percentage points, that removing band fusion reduces accuracy to 9.2%, and that Gamma outperforms Mu and Beta as a standalone band feature.

Synthesis

The following synthesis is organised around three themes derived from the primary research question. Each theme draws across the full set of nine included studies to identify patterns, convergences, and gaps in the evidence. The fourth theme, governance implications, is addressed in the discussion section where its interpretive weight is more appropriately carried.

The Cross-Subject Performance Landscape

Reported cross-subject accuracy figures across the nine included studies span a considerable range, and understanding what that range reflects requires attending carefully to the conditions under which each result was produced. Differences in task complexity, evaluation protocol, and the availability of target subject data at test or adaptation time each contribute to the variation in reported figures, and disaggregating these sources of variance is a necessary step before any cross-study comparison can be meaningfully drawn.

The studies employing the most explicitly described and methodologically detailed LOSO protocols - Saha and Fels (2019), Jiménez-Guarneros and Gómez-Gil (2021), and Ng and Guan (2024) - produced results that, while varying in absolute magnitude, reflect a consistent relationship between task complexity and achievable cross-subject accuracy. Saha and Fels (2019) reported the highest verified cross-subject figures in the included set, with LOSO Phase Two accuracy ranging from approximately 73% to 85% across five phonological tasks, each

framed as a binary classification problem with a chance level of 50%. When the same laboratory's framework is extended to 11-class speech token identification in Saha et al (2019), the reported figure of 53.36% is generated from a pooled random split rather than a complete LOSO evaluation, meaning it is not directly comparable with the binary LOSO results from the prior paper. The 11-class LOSO confusion matrix in that study - which shows a shift from near-random scattering under raw features to a recognisably diagonal structure under the hierarchical phonological framework - provides informative cross-subject evidence, though it is not accompanied by a single summary accuracy figure that would allow direct placement alongside results from the other included studies.

On the Thinking Out Loud dataset, three studies evaluate the same four-class task against a 25% chance level, providing the closest approximation to a shared benchmark available in this literature. Radwan et al. (2024) reported 32% from a single stratified split without cross-validation. Ng and Guan (2023) reported 29.92% under implied subject-level holdout. Ng and Guan (2024) reported 25.55% under a verified zero-shot LOSO condition and 31.15% under a few-shot condition supplying 50 labeled target trials. These figures are broadly consistent with one another and suggest that cross-subject accuracy on this dataset, under conditions approximating genuine subject independence, falls within a narrow band of approximately five to seven percentage points above chance.

Guo et al. (2024) reported 32.2% on a five-class task against a 20% chance level, representing a 12.2 percentage point margin above chance - the widest such margin in the included set for any multi-class task evaluated under LOSO-style conditions. This result, however, is achieved with 80% of the target subject's labeled covert EEG available for fine-tuning. The corresponding zero-shot figure of 24.5%, reported within a few-shot analysis

rather than as the primary result, reduces the above-chance margin to 4.5 percentage points, placing it within the same performance band as the Thinking Out Loud results when calibration data is removed.

The figure that sits furthest from the rest of the included set is TeleProNet's 93.84% on the 11-class Kara One task reported by Zhang et al (2025). As documented in the descriptive summary, this result is produced under an episodic prototypical classification scheme in which 33 labeled trials from the held-out subject are used at inference time to construct class prototypes. The architecture requires this labeled support set to function, meaning the result is not directly comparable with studies that evaluate under zero-shot or purely subject-independent conditions. Bhalerao and Pachori (2025) report 60.80% on a six-class task, though as noted in the descriptive summary the within-dataset evaluation protocol does not confirm subject-level holdout in sufficient detail to treat this as a fully verified cross-subject result under this review's operational definition.

What emerges from examining the cross-subject performance landscape across the nine studies is that the results most closely approximating genuine zero-shot generalisation - where no target subject data of any kind is involved and subject-level partitioning is explicitly verified - consistently fall in a range of moderate above-chance accuracy on tasks of three or more classes. Studies reporting substantially higher figures tend to either target binary tasks, involve some form of target subject data at adaptation or inference time, or employ evaluation protocols whose subject-level partitioning cannot be fully confirmed from the reported methodology. This is not a criticism of any individual study's contribution; the approaches vary in their aims and many are deliberately designed to explore calibration-assisted adaptation rather than zero-shot

generalisation. It is, however, a structural observation about what the current evidence base can and cannot support as a claim about cross-subject decoding capability.

The Calibration Burden

The amount of target subject data a system requires before it can produce meaningful classifications is a dimension of cross-subject generalisation that runs alongside reported accuracy as a determinant of practical deployability. Across the nine included studies, this requirement varies substantially, is reported with differing degrees of transparency, and is framed in ways that do not always make the practical implications immediately visible to the reader. Mapping these requirements across the set reveals a consistent empirical pattern: reported performance tends to increase with the quantity of target subject data made available, and the studies that report the strongest results are, with few exceptions, also those that involve the greatest calibration investment.

Three studies in the included set report results that require no target subject data at inference or adaptation time. Saha and Fels (2019) and Saha, Abdul-Mageed and Fels (2019) both operated under fully subject-independent conditions, with the test subject's data withheld entirely throughout both training and evaluation phases. Bhalerao and Pachori (2025) similarly reported no target subject fine-tuning requirement, though as noted earlier the within-dataset evaluation protocol introduces some uncertainty about the strictness of subject-level separation.

Jiménez-Guarneros and Gómez-Gil (2021) occupied a distinct position in the calibration spectrum. The proposed method requires no labeled data from the target subject, which the authors describe as a fully unsupervised adaptation approach. The adaptation phase does, however, use 80% of the target subject's unlabeled EEG to re-estimate batch normalisation statistics and refine the conditional distribution of the learned feature space, with the remaining

20% reserved for testing. The distinction between labeled and unlabeled target data is methodologically meaningful, and the authors are transparent about this design. At the same time, the practical implication is that a recording session of substantial duration from the target subject is required before the system can be adapted, which represents a data collection demand that is worth noting alongside the labeled-versus-unlabeled distinction.

Ng and Guan (2023) and Ng and Guan (2024) both tested subject-adaptive variants that supply 50 labeled trials from the target subject for fine-tuning, and both report that this condition produces the best results in their respective evaluations. Ng and Guan (2024) is the only study in the included set that treats the calibration quantity itself as a primary experimental variable, reporting performance across a gradient from zero to 50 labeled trials and finding a consistent improvement with each increment. The zero-shot inner speech result of 25.55% and the 50-trial result of 31.15% together provide the clearest empirical illustration in the included literature of the relationship between calibration volume and classification performance on this task.

Guo et al. (2024) involved the largest absolute quantity of labeled target subject data in the included set, using approximately 80 labeled trials from each target subject for fine-tuning. The zero-shot condition is included in the analysis and reported at 24.5%, though it appears within a few-shot exploration table rather than as a primary comparative result. Zhang, Ding and Chen (2025) required 33 labeled trials from the test subject at inference time as a structural feature of the prototypical classification scheme, since class prototypes cannot be constructed without labeled support examples from the target. This requirement is not framed as a calibration burden in the paper but follows directly from the architecture's design.

Across the nine studies, no study that reports its best results under a zero-shot or fully subject-independent condition, without any form of target subject data, exceeds 85% accuracy,

and these results are restricted to binary tasks. On tasks of four or more classes, zero-shot cross-subject accuracy across the included set does not exceed approximately 32% in any study with a verifiable subject-level partition. The studies that approach or substantially exceed this threshold do so with target subject data contributing to the result, whether through labeled fine-tuning, unlabeled adaptation, or labeled inference-time prototype construction. This pattern across the evidence base suggests that calibration, understood broadly as any requirement for target subject data prior to or during deployment, is not an optional enhancement in current cross-subject imagined speech decoding systems but a structural condition on which their reported performance substantially depends.

Methodological Fragmentation

One of the more consequential structural features of the included literature is the degree to which evaluation standards, terminology, dataset usage, and reporting practices vary across studies in ways that make direct comparison difficult and cumulative progress hard to assess. The nine included studies were selected precisely because they address cross-subject generalisation, yet they do so using different datasets, different definitions of what subject independence requires, different metrics, and different levels of reporting detail. This fragmentation is not unique to any individual study but appears as a field-level characteristic of the literature at this stage of development.

Dataset concentration is one observable dimension of this fragmentation. The Kara One dataset appears in three of the nine included studies, and the Thinking Out Loud dataset appears in three others, meaning six of nine studies draw on one of just two datasets. The remaining three studies use datasets that do not appear elsewhere in the included set. While the reuse of established public datasets is a reasonable and often necessary practice in a field where data

collection is resource-intensive, the concentration of findings around two datasets limits the extent to which results can be treated as independent contributions to a cumulative evidence base. The three Kara One studies differ in their subject retention decisions, one uses all 14 subjects, another uses 14, and the third retains only 8 after quality filtering, which means that even within the same dataset, results are not generated on comparable subject pools.

Evaluation terminology presents a related challenge. Several of the included studies use the terms cross-subject, subject-independent, and generalisation interchangeably or without operational definition, while describing evaluation procedures that differ from one another in ways that materially affect what the reported accuracy figures represent. Leave-one-subject-out cross-validation is the most commonly stated framework, appearing in seven of the nine studies, but the procedural descriptions accompanying this label vary from highly detailed and verifiable accounts to brief mentions without specification of how subjects were partitioned across training, validation, and test roles, how many repetitions were conducted, or how hyperparameters were selected in relation to the subject-level split. In two studies, LOSO is applied selectively, to specific comparison tables or visualisations, rather than as the primary evaluation scheme, while pooled or randomly split results are reported as the headline figures. These variations do not necessarily reflect methodological errors in individual studies, but they do mean that a reader comparing accuracy figures across the set is not comparing quantities generated under equivalent conditions.

The treatment of statistical validation follows a similarly uneven pattern. Three studies — Jiménez-Guarneros and Gómez-Gil (2021), Ng and Guan (2024), and Guo et al. (2024) - accompanied their cross-subject results with formal significance testing. The remaining six report accuracy figures, and in several cases standard deviations, without statistical testing of any

kind. In a domain where subject counts are typically small and per-subject variance is high, the interpretability of a reported mean accuracy without a confidence interval or significance test is limited, and the absence of statistical validation across a majority of the included studies is a recurring feature of the evidence base rather than an isolated gap.

Reporting of standard deviations is inconsistent in a related way. Six studies report standard deviations alongside mean accuracy, providing at least a partial indication of cross-subject variability. Three do not. Where standard deviations are reported, the range across subjects is in several cases substantial - Guo et al. (2024) reported a standard deviation of plus or minus 9.1 percentage points on the headline result, and Zhang et al (2025) reported plus or minus 6.03% on a mean of 93.84% - indicating that individual subject performance varies considerably around the reported mean. The absence of standard deviations in the remaining studies means that the degree of cross-subject variability in those results cannot be assessed from the reported figures alone.

Feature extraction and preprocessing choices add a further layer of heterogeneity that compounds the difficulty of cross-study comparison. The included studies employ codebook-based histogram representations, cross-covariance matrices, multivariate joint time-frequency images, spatiotemporal VAE latent features, multi-band Hilbert envelopes, end-to-end learned CNN representations, and raw EEG without explicit feature extraction. Preprocessing pipelines vary in bandpass filter ranges, artifact removal methods, channel selection procedures, and epoch windowing strategies. These choices are each defensible within individual study designs, and several are motivated by principled neurophysiological reasoning. Taken together, however, they mean that reported performance differences across studies reflect a compound of architecture, features, preprocessing, and evaluation protocol, making it difficult

to isolate the contribution of any single methodological decision to cross-subject generalisation outcomes.

The class vocabulary across the included studies similarly varies in ways that affect comparability. Tasks range from binary phonological contrasts to 4-class directional word sets to 5-class command phrases to 11-class phoneme-and-word combinations. Chance levels correspondingly range from 50% to approximately 9%, and a percentage point of accuracy gain represents a very different proportion of the available performance space depending on the number of classes. Studies are not always explicit about reporting results relative to the task-specific chance level, which can make absolute accuracy figures appear more or less impressive depending on the reader's familiarity with the task structure.

Discussion

This review set out to answer two research questions: what the current state of cross-subject generalisation in EEG-based covert speech decoding is, and what governance and ethical implications follow from the identified challenges. The synthesis established the empirical picture: performance on a common four-class benchmark clusters narrowly above chance regardless of generalisation strategy, higher figures in the literature reflect task simplification or calibration dependency rather than zero-shot advances, and the field is characterised by methodological fragmentation that makes cumulative progress difficult to track. This discussion takes that picture as its starting point and asks what it means, for the field's research priorities, for its governance readiness, and for the populations it claims to serve.

The Generalisation-Calibration Tradeoff

The central finding of this review is not that cross-subject generalisation in EEG-based imagined speech decoding is impossible, but that it is not yet separable from calibration in the

way the literature's framing often implies. Across the nine included studies, reported accuracy figures vary substantially, but that variation tracks closely with two variables that are rarely foregrounded in how results are presented: the complexity of the classification task and the quantity of target subject data made available before or during evaluation. When these two variables are held constant, that is, when attention is restricted to results generated under verified zero-shot conditions on tasks of three or more classes, the performance range across the included set narrows considerably, converging on figures that are statistically above chance but modest in absolute terms. This convergence is the most important empirical observation this review can offer, and it deserves to be stated plainly before its implications are drawn out.

The tradeoff itself can be described as follows. A system that requires no target subject data at all, no labeled trials for fine-tuning, no unlabeled recordings for distribution alignment, no support examples for prototype construction, operates at the lower end of the performance range observed across the included studies on tasks of meaningful complexity. A system that incorporates target subject data in any of these forms moves up that performance range, sometimes substantially, but in doing so reintroduces a requirement that the intended beneficiary of the technology must actively contribute data before the system can serve them. These two properties, generalisation capability and calibration independence, pull in opposite directions in the current evidence base, and no included study achieves both simultaneously at a level that would constitute clinically meaningful performance on a multi-class task.

This tradeoff has not, to this review's knowledge, been named explicitly as a unified constraint in the imagined speech BCI literature, though its constituent observations are present across multiple included studies. The consequence of leaving it unnamed is that the two sides of the tradeoff are addressed in parallel research streams, transfer learning and domain adaptation

on one side, zero-shot and meta-learning approaches on the other, without a shared vocabulary for discussing what is being traded between them. A field that does not name the tradeoff it is navigating would be less likely to make deliberate progress on it, and the methodological fragmentation documented in the synthesis is in part a symptom of this.

It is worth being clear about what this observation does and does not imply. It does not imply that the approaches documented in the included studies are misguided or that the performance figures they report are uninformative. Many of the included studies are precisely calibrated to explore calibration-assisted adaptation, and the question of how much target subject data is needed to achieve a given performance level is a legitimate and practically important research question. What the observation does imply is that results generated under calibration-assisted conditions should be understood as answering a different question than results generated under zero-shot conditions, and that presenting both under the same label of cross-subject generalisation, without distinguishing between them, makes the evidence base harder to read than it needs to be. Establishing a shared reporting convention that makes the calibration condition explicit alongside the accuracy figure would, by itself, significantly improve the interpretability of future work in this space.

Governance and Ethical Implications

What the Evidence Can and Cannot Support

A recurring feature of public and policy discourse around brain-computer interfaces is the invocation of imagined speech decoding as a technology that is either imminently capable of reading minds or already approaching that threshold (BBC Future, 2026; Mullin, 2026). The evidence assembled in this review does not support that characterisation for EEG-based systems, and being precise about what the evidence does and does not show is itself a governance

contribution, since policy responses calibrated to an overstated threat model risk misallocating regulatory attention.

The findings of this review establish several concrete boundaries on current capability. First, verified zero-shot cross-subject decoding of imagined speech from EEG, meaning classification of inner linguistic content from an individual whose data was entirely absent from training, without any recording session or labeled examples from that individual, does not exceed modest above-chance accuracy on tasks of meaningful complexity in any included study. The highest verified zero-shot result on a multi-class task in this review is 32% on a four-class problem, representing a margin of seven percentage points above chance. Second, the calibration requirement that produces stronger results is not a minor tuning step but a substantive data collection process: the studies reporting the strongest calibration-assisted results require between 33 and approximately 80 labeled trials from the target subject, collected under controlled experimental conditions. Third, all included studies recruit healthy adult participants under laboratory conditions, and none demonstrates deployment-ready performance on the clinical populations, individuals with ALS, locked-in syndrome, or severe motor impairment, that the technology is most commonly invoked as serving.

Together, these boundaries suggest that the scenario underlying the most acute public concern about imagined speech BCIs, that a third party could covertly decode an individual's inner speech without that individual's knowledge or participation, may not be cleanly or fully supported by the current state of the evidence for EEG-based systems. The calibration requirement alone constitutes a practical barrier to covert decoding: a system that needs labeled examples from a specific individual before it can classify that individual's neural signals cannot be deployed against an unconsenting, uncalibrated person. This does not mean that privacy

concerns in this space are without foundation, but it does mean that the specific concern about covert involuntary mind reading, at least for EEG-based imagined speech systems at their current capability level, reflects a threat model that outruns the technical evidence.

Acknowledging this gap between discourse and evidence is not a dismissal of governance concern. It is, rather, a precondition for directing that concern appropriately, toward the risks that are present now, under current capability, rather than those that remain speculative.

Concerns That Do Not Await Further Maturity

Even at current capability levels, the research practices and data collection infrastructure documented in this review give rise to governance concerns that are not contingent on future advances in decoding performance. These concerns fall into three overlapping categories: the sensitivity of neural data as a category of personal information, the structural gap between the populations studied and the populations the technology claims to serve, and the consent architecture implied by systems that require active participant contribution.

Neural data occupies a distinctive position in the landscape of personal information. EEG recordings collected for imagined speech research encode not only the target signal, the neural correlates of specific imagined words, but the broader electrical activity of the participant's brain across the recording session, which may include incidental cognitive content beyond what the participant intends to produce (Ienca & Andorno, 2017). Unlike most biometric data, neural signals cannot be easily anonymised: individual EEG signatures carry identifying properties that persist across sessions and contexts, and research has demonstrated that EEG recordings can be used for biometric identification of individuals with meaningful accuracy (Kumar et al., 2017; Li et al., 2022; Magee et al., 2024; Zhang et al., 2018). This means that a dataset of imagined speech EEG recordings is simultaneously a dataset of biometric identifiers, and the data

protection implications of that dual character apply independently of whether the decoding models trained on those recordings perform well or poorly.

The six datasets used across the nine included studies vary considerably in their availability: several are publicly accessible, meaning that the raw neural recordings of the participants who contributed to those datasets are downloadable by any researcher with institutional access. The data protection frameworks applicable to this practice, including the General Data Protection Regulation in the European context, which classifies data concerning health and biometric identification as special category data requiring heightened protection (GDPR, 2016, Art. 9), were not designed with the specific properties of neural data in mind, and whether existing frameworks provide adequate protection for EEG datasets of this kind is a question that legal and regulatory scholarship has begun to engage with but has not resolved (Ienca, 2021). The finding that no included study contains a substantive discussion of data sensitivity or storage governance is therefore not merely an observation about individual research teams but a signal about a field-level gap in how neural data stewardship is being conceptualised.

The structural exclusion of clinical populations from the included studies raises a related but distinct concern. Seven of the nine included studies invoke individuals with ALS, locked-in syndrome, or severe motor impairment as the intended beneficiaries of imagined speech BCI technology in their introductions, but none recruits participants from these populations, discusses deployment constraints specific to them, or acknowledges the gap between the healthy adult populations studied and the clinical contexts motivating the research. This gap matters for governance in a concrete way: the calibration requirements documented across the included studies, which range from 33 to approximately 80 labeled trials collected under controlled laboratory conditions, are described and evaluated in relation to healthy adult participants who

can follow instructions, sustain attention, and produce reliable neural signals across extended recording sessions. The feasibility of equivalent calibration procedures for individuals with severe motor impairment, cognitive fatigue from progressive neurological disease, or limited capacity to sustain experimental protocols is an entirely separate question that the current evidence base does not address. A technology whose deployment scenario involves populations systematically excluded from its evaluation carries a particular obligation to engage with this gap (Finkelstein, 2025) and the absence of that engagement in the included literature is a governance-relevant observation regardless of where decoding performance currently stands.

Consent architecture is a third area where current practice warrants attention. Imagined speech decoding systems, even at their current capability level, process neural signals that encode inner linguistic content - content that participants produce intentionally as part of the experimental task but that may also include incidental cognitive activity beyond the target. The concept of informed consent in neuroscience research is well established (O'Shaughnessy et al., 2023) but its application to systems that capture a continuous stream of neural activity, including content the participant may not intend to produce, involves questions about the boundaries of what a participant can meaningfully consent to in advance (Hendriks et al., 2019). These questions are not resolved by existing consent frameworks and are not engaged within any of the nine included studies. Raising them is not a claim that current research practice is ethically deficient; it is a recognition that the consent frameworks adequate for traditional neuroimaging studies may need to be revisited as decoding systems become more capable of extracting unintended content from neural recordings.

Neurorights frameworks, which have emerged in legal and philosophical scholarship as a proposed extension of human rights doctrine to cover the neural substrate of cognition, are

relevant here as a horizon rather than an immediate regulatory instrument (Cornejo-Plaza et al., 2024; Ienca, 2021). Chile's 2021 constitutional amendment establishing neurorights and the neurorights legislation introduced in several US states represent early attempts to give legal form to these principles, though the specific protections they afford and the technical thresholds at which they become operative remain subjects of active debate. For EEG-based imagined speech decoding specifically, the calibration dependency documented in this review suggests that the most acute neurorights concerns, those premised on nonconsensual decoding of inner thought, are not yet technically grounded for this modality, while the more foundational concerns about data sovereignty, the right not to have neural data collected without meaningful consent, and the right to cognitive privacy in a broad sense apply at current capability levels and do not require further advances in decoding performance to be actionable.

The Distance Between Current Practice and Clinical Translation

Beyond the data and consent concerns that apply at current capability levels, the methodological fragmentation documented in the synthesis carries its own governance implication: a field whose results cannot be independently verified, compared across studies, or reproduced from reported methodology is not in a position to support the kind of regulatory scrutiny that clinical translation would require.

Regulatory frameworks for AI-based medical devices and clinical decision support systems, including the EU AI Act's classification of certain AI systems as high-risk where they are intended to influence clinical outcomes (EU AI Act, 2024, Annex III), and the FDA's evolving guidance on software as a medical device (Abram et al., 2026), share a common expectation: that the performance claims made for a system are verifiable, that the evaluation conditions under which those claims were generated are transparent, and that the system behaves

consistently across the population it is intended to serve. The evidence base assembled in this review does not currently meet these expectations in several respects. Evaluation protocols are inconsistently described, calibration requirements are inconsistently reported, statistical validation is absent in the majority of included studies, and the subject pools on which performance claims are based are drawn exclusively from healthy adult populations rather than the clinical populations for whom deployment is envisioned.

This does necessarily imply that regulatory engagement with imagined speech BCI is premature. It means, rather, that the regulatory readiness of this technology as a clinical tool depends not only on advancing decoding performance but on establishing the reporting and evaluation standards that would allow performance claims to be assessed with the confidence that clinical regulation requires. The EU AI Act's transparency and documentation requirements, which mandate that high-risk AI systems be accompanied by technical documentation sufficient to allow conformity assessment (EU AI Act, 2024, Arts. 11, 13), provide a useful external standard against which to assess what the field would need to produce before a cross-subject imagined speech decoding system could plausibly enter a regulatory pathway. By that standard, the current literature has considerable distance to travel, not because the science is inadequate, but because the reporting infrastructure that would allow the science to be evaluated has not yet been established as a field norm.

Limitations

This review was conducted by a single reviewer, which introduces the possibility of screening bias at both the title and abstract and full-text stages that independent dual-reviewer screening would reduce. The keyword-based exclusion heuristic applied at the title and abstract stage, excluding records with no cross-subject signal in title, abstract, or keywords, was validated

post-hoc through a random sample of 25 excluded records, all of which were confirmed as not meeting the operational definition of cross-subject generalisation. This provides partial but not complete reassurance that eligible studies were not excluded at this stage.

The operational definition of cross-subject generalisation applied in this review - requiring strict subject-level partition between training and test data with no contribution from the test subject to model training, hyperparameter selection, or feature extraction - is more restrictive than the definitions implicitly or explicitly applied in some of the included studies. This strictness is deliberate and motivated by the primary research question, but it means that the review's conclusions about verified cross-subject performance are restricted to a subset of the evidence that meets this definition, and may not generalise to the broader literature on cross-subject methods that this review's screening did not capture.

The heterogeneity of methods, datasets, evaluation protocols, and reporting practices across the nine included studies precluded quantitative synthesis, and the narrative synthesis presented here reflects interpretive judgments made by a single reviewer. The thematic structure of the synthesis, particularly the framing of the generalisation-calibration tradeoff, represents an analytical lens developed inductively from the evidence rather than a pre-specified framework, and alternative thematic organisations of the same evidence are possible.

Finally, the governance analysis in this review draws on publicly available neuroethics and regulatory literature and reflects the state of that literature as of early 2026. The regulatory landscape for AI-based neurotechnology is evolving rapidly, and specific frameworks cited here may have been updated or superseded by the time this review is read.

Delimitations

This review is deliberately bounded by several scope decisions. First, EEG was selected as the sole signal acquisition modality of interest; studies employing fMRI, MEG, fNIRS, ECoG, or multimodal combinations as the primary recording approach were excluded, reflecting a focus on the non-invasive, portable modality most relevant to real-world deployment.

Second, the review is restricted to imagined, inner, covert, and silent speech paradigms, excluding overt speech production and speech perception tasks, which engage meaningfully different neural processes and decoding pipelines.

Third, only studies that addressed cross-subject generalization as defined operationally in this review were included; the substantially larger literature on subject-dependent decoding was outside scope.

Fourth, the search was confined to English-language publications from 2014 to February 2026, reflecting both the period of meaningful machine learning application to this domain and practical resource constraints.

Finally, the governance and ethical analysis in this review is interpretive and grounded in the synthesized technical evidence rather than constituting a standalone legal or regulatory analysis; questions of precise legislative applicability were outside the scope of what this review set out to establish.

Methodological Recommendations

The methodological fragmentation documented across the included studies suggests several concrete areas where field-level standardisation would improve the interpretability and cumulativeness of future work on cross-subject imagined speech decoding.

The most consequential single change would be the adoption of a shared reporting convention that makes the calibration condition explicit alongside every reported accuracy

figure. A result generated under zero-shot conditions, a result generated with unlabeled target subject data for adaptation, and a result generated with labeled target subject data for fine-tuning are not the same quantity and should not be reported under the same label. A minimal reporting standard might require that any cross-subject accuracy figure be accompanied by a clear statement of how much target subject data, if any, was used to produce it, and in what form.

Evaluation protocols should be described in sufficient procedural detail to allow independent verification that subject-level partitioning was maintained throughout training, hyperparameter selection, and evaluation. Where leave-one-subject-out cross-validation is used, the role of each subject subset, training, validation, and test, should be specified, and the number of repetitions and random initialisations reported. Statistical significance testing and confidence intervals should accompany cross-subject results as a default rather than an exception, given the small subject counts and high per-subject variability characteristic of this literature.

Dataset diversity beyond the Kara One and Thinking Out Loud collections is needed to establish whether findings generalise across recording equipment, electrode configurations, stimulus modalities, and linguistic systems. The concentration of findings around two datasets limits the independence of the evidence base in ways that are difficult to remedy within individual studies but could be addressed through coordinated data collection efforts or cross-dataset evaluation as a standard component of study design.

Recruitment of participants from clinical populations, or at minimum, explicit engagement with the question of how reported findings would generalise to those populations, is necessary for the research to maintain coherence between its stated motivation and its actual evidence base. This does not require that every study recruit participants with ALS or locked-in

syndrome, but it does require that the gap between studied and intended populations be acknowledged and addressed rather than left implicit.

Future Research Directions

The generalisation-calibration tradeoff identified in this review points to several directions where progress would constitute genuine advancement toward the clinical deployment scenarios that motivate this field.

The development of evaluation benchmarks that separately track zero-shot generalisation performance and calibration efficiency, quantifying not only how well a system performs with a given quantity of target subject data but how rapidly performance improves as calibration data is added, would provide a more informative and actionable picture of system capability than single accuracy figures under fixed conditions. Meta-learning approaches, which explicitly optimize for rapid adaptation from small amounts of target data, are promising in this regard and warrant further investigation on larger and more diverse subject pools than are currently available.

Progress on the data scarcity problem that underlies the calibration burden requires investment in larger, more diverse, and more openly shared EEG datasets for imagined speech, including datasets that recruit clinical populations and that standardise stimulus design, preprocessing pipelines, and electrode configurations sufficiently to support cross-dataset evaluation. The concentration of the current literature around a small number of datasets is in part a function of the resource demands of EEG data collection, and coordinated efforts to expand the available data infrastructure would have a disproportionate impact on the field's ability to make and verify generalisation claims.

The privacy and data governance questions raised in this review point toward research directions at the intersection of neural signal processing and privacy-preserving machine

learning. Techniques such as federated learning, differential privacy, and on-device processing have been explored in adjacent BCI domains (Ghader et al., 2023; Liu et al., 2024) and represent a direction where technical and governance considerations can be addressed jointly rather than sequentially.

Conclusions

This review set out to characterise the current state of cross-subject generalisation in EEG-based imagined speech decoding and to examine its implications for governance and clinical translation. The evidence assembled across nine included studies supports a picture that is more constrained than headline accuracy figures might suggest, and more nuanced than either optimistic deployment narratives or alarmist mind-reading framings would imply.

The central analytical contribution of this review is the identification of the generalisation-calibration tradeoff as a structural feature of the current evidence base: cross-subject decoding performance and calibration independence are not independent properties that can be optimised separately, but linked quantities that trade against each other in ways that current approaches have not fully resolved. Recognising this tradeoff explicitly is a precondition for making deliberate progress on it, and for communicating honestly about what current systems can and cannot do.

The governance analysis that follows from this finding is deliberately calibrated to the evidence. The concerns that are most salient in public discourse about imagined speech BCIs, covert, nonconsensual decoding of inner thought, are not supported by the current technical evidence for EEG-based systems, where the calibration requirement alone constitutes a practical barrier to the scenarios that underlie those concerns. The concerns that are warranted now, at current capability levels, are less dramatic but more tractable: the sensitivity of neural data as a

category of personal information, the structural gap between the populations studied and those the technology claims to serve, the adequacy of existing consent frameworks for systems that capture continuous neural activity, and the distance between current reporting practices and the transparency standards that clinical regulation would require.

A field that addresses these concerns seriously, not because deployment is imminent, but because the research infrastructure being built now will shape what deployment looks like when it becomes possible, is in a stronger position both scientifically and ethically than one that defers governance to a later stage of maturity. That is the practical import of this review's findings, and it is offered in that spirit.

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